12th

Wednesday, September 23, 2026 9:55 AM

Beam equation

$$\begin{cases} u^{(iv)} = f \quad \text{in } (0,1) \\ u(0) = u(1) = u'(0) = u'(1) = 0 \end{cases}$$

$$V = \left\{ v : \int_0^1 \left[v^2 + (v')^2 + (v'')^2 \right] dx < \infty \right.$$
$$\left. v(0) = v(1) = v'(0) = v'(1) = 0 \right\}$$

key functions in V are C^1

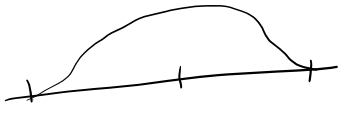
find $u \in V$ such that

$$\int_0^1 u'' v'' \, dx = \int_0^1 f v \, dx$$

$$\forall \; v \in V$$

FEM: $V_h \subset V$

$\phi_j^{(1)}$:



$\phi_j^{(2)}$:



$$U = \sum_{j=1}^{M} \xi_j \phi_j^{(1)}(x) + \sum_{j=1}^{M} \eta_j \phi_j^{(2)}(x)$$

if $u \in V$ then

$$\Pi_h u = \sum_{j=1}^{m} u(x_j)\phi_j^{(1)}(x) + \sum_{j=1}^{n} u'(x_j)\phi_j^{(2)}(x)$$

where, $\Pi_h u \in V_h$

Interpolation errors:

$$\|u - \Pi_h u\|_{L^2(0,1)} \le c_1 \|h^4 u^{(4)}\|_{L^2(0,1)}$$

$$\|(u - \Pi_h u)'\| \le c_2 \|h^3 u^{(4)}\|_{L^2(0,1)}$$

$$\|(u - \Pi_h u)''\|_{L^2(0,1)} \le c_3 \|h^2 u^{(4)}\|_{L^2(0,1)}$$

ex (uniform mesh)

$h_j = h \quad \forall j$

$$\int_0^1 u'' v'' dx = \int_0^1 f v dx$$

where $v = \phi_1^{(1)}, \dots, \phi_m^{(1)}, \phi_1^{(2)} \dots \phi_m^{(2)}$

Stiffness Matrix:

$$\int_0^1 u'' v'' dx = \sum_{j=1}^{m} \xi_j \int_0^1 (\phi_j^{(1)})'' (\phi_k^{(1)})'' dx$$

$$\int_0^1 v'' v'' \, dx = \sum_{j=1}^{m} \xi_j \int_0 (\phi_j^{(1)})'' (\phi_k^{(1)})'' \, dx$$

$$+ \sum_{j=1}^{m} \eta_j \int_0^1 (\phi_j^{(2)})'' (\phi_k^{(1)})'' \, dx$$

$k = 1, 2, \cdots m$

then again

$$\sum_{j=1}^{m} \xi_1 \int_0^1 (\phi_j^{(1)})'' (\phi_k^{(2)})'' \, dx$$

$$+ \sum_{j=1}^{m} \eta_j \int_0 (\phi_j^{(2)})'' (\phi_k^{(2)})'' \, dx$$

$k = 1, 2, \cdots m$

$$\begin{bmatrix} \underline{\underline{A}} & \underline{\underline{C}} \\ -\underline{\underline{C}} & \underline{\underline{B}} \end{bmatrix} \in \mathbb{R}^{2m \times 2m}$$

Here $\underline{\underline{A}}, \underline{\underline{B}}, \underline{\underline{C}} \in \mathbb{R}^{m \times m}$

Now,

$$A_{jk} = \int_0^1 (\phi_j^{(1)})'' (\phi_k^{(1)})'' \, dx$$

$$\underline{\underline{A}} = \frac{12}{h^3}\begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & & & -1 \\ & & & -1 & 2 \end{bmatrix}$$

$m \times m$, tridiagonal, symmetric

$$B_{jk} = \int_0^1 \left(\phi_j^{(2)}\right)'' \left(\phi_k^{(2)}\right)'' dx$$

$$\underline{\underline{B}} = \frac{2}{h}\begin{bmatrix} 4 & 1 & & & \\ 1 & 4 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & & & 1 \\ & & & 1 & 4 \end{bmatrix}$$

Now

$$C_{jk} = \int_0^1 \left(\phi_j^{(1)}\right)'' \left(\phi_k^{(2)}\right)'' dx$$

$$\underline{\underline{C}} = \frac{6}{h^2}\begin{bmatrix} 0 & 1 & & & \\ -1 & \ddots & \ddots & & \\ & \ddots & \ddots & 1 & \\ & & -1 & 0 & 1 \\ & & & -1 & 0 \end{bmatrix}$$

$m \times m$ tridiagonal (Not symmetric

$M \times M$ tridiagonal (Not symmetric

anti-symmetric $\underline{\underline{C}}^T = -\underline{\underline{C}}$

$$\therefore \underline{\underline{S}} = \left[\begin{array}{c|c} \underline{\underline{A}} & \underline{\underline{C}} \\ \hline \underline{\underline{C}}^T & \underline{\underline{B}} \end{array}\right]$$

$$\underline{\underline{S}}^T = \underline{\underline{S}}$$

$\therefore$ S is $2M \times 2M$, symmetric.

R.H.S : $f \longrightarrow \tilde{\pi}_h f = \sum_{j=0}^{m+1} f(x_j)\phi_j^{(1)}(x)$

$$+ \sum_{j=0}^{m+1} f'(x_j)\phi_j^{(2)}(x)$$

$$\int_0^1 f v \, dx \longrightarrow \int_0^1 \left(\tilde{\pi}_h f\right) v \, dx$$

Now,

$$f_j^{(1)} = \frac{h}{70}\left[9f(x_{j-1}) + 52f(x_j) + 9f(x_{j+1})\right]$$

$$+ \frac{13h^2}{420}\left[f'(x_{j-1}) - f'(x_{j+1})\right]$$

$j = 1, 2, \cdots M$

420

$j = 1, 2, \cdots m$

Now

$$f_j^{(4)} = \frac{13h^2}{420}\left[f(x_{j+1}) - f(x_{j-1})\right] - \frac{h^3}{420}\left[3f'(x_{j-1}) - 8f'(x_j) + 3f'(x_{j+1})\right]$$

$j = 1, 2, \cdots m$

$\therefore$ final $2M \times 2M$ system:

$$\begin{bmatrix} A & C & & \\ C & B & C & \\ & C & B & C \\ & & \ddots & \ddots \end{bmatrix} \begin{bmatrix} z \\ \vdots \\ z \end{bmatrix} = \begin{bmatrix} f^{(1)} \\ \vdots \\ f^{(4)} \end{bmatrix}$$

z ← derivative value at node

$2M \times 2M$, symmetric, positive definite.